

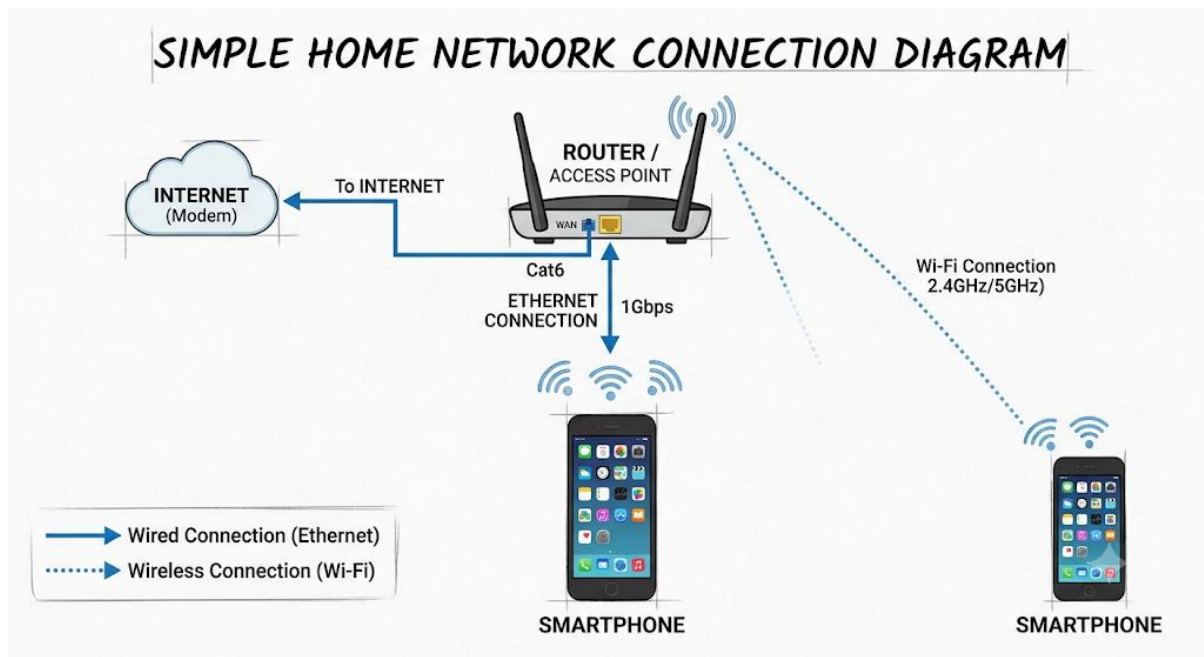
Wireless Networking

Task (1):

Three Popular Wi-Fi standards:

- 802.11n (Wi-Fi 4): Introduced dual-band operation on both 2.4 GHz and 5 GHz frequencies, offering maximum theoretical data rates of up to 600 Mbps.
- 802.11ac (Wi-Fi 5): Operates exclusively on the 5 GHz band for consumer access, delivering a significant leap in throughput with theoretical speeds reaching up to 3.5–6.9 Gbps by utilizing wider channels (up to 80/160 MHz).
- 802.11ax (Wi-Fi 6 / 6E): Expands spectrum support to include the 6 GHz band (in Wi-Fi 6E) alongside 2.4 GHz and 5 GHz, pushing theoretical speeds up to 9.6 Gbps with improved efficiency in congested, multi-device environments.

Task (2):



Task(3):

No.	SSID (Wi-Fi Name)	Signal Strength	Security Type	Choice
1	Home_WiFi	Excellent (90%)	WPA3	☒ Choose
2	College_WiFi	Good (75%)	WPA2	—
3	Neighbor_WiFi	Weak (35%)	WPA2	—

Task (4):

SSID hiding

SSID hiding means configuring your router to not broadcast its network name publicly, so it won't show up in the list of available Wi-Fi networks that nearby devices scan for — anyone wanting to connect has to already know the exact SSID and type it in manually.

- **Advantage:** It adds a small layer of obscurity — casual, non-technical passersby won't even see your network exists, which can reduce opportunistic connection attempts.
- **Disadvantage:** It doesn't provide real security — anyone using basic Wi-Fi scanning tools can still detect the hidden network's presence, and it actually makes connecting your own legitimate devices more annoying, since you have to manually enter the SSID each time instead of just picking it from a list.